

Bank Account Test Plan

Introduction

- Project Name: Bank Account System
- Author: Vince N.
- Date: May 2026
- Version: 1.0
- Description: A simple banking system that allows user to log in, deposit money, and withdraw money

Scope

In Scope	Out of Scope
UserLogin authentication	Database Testing
UserLogin input validation	UI testing
BankAccount deposit functionality	Performance Testing
BankAccount withdrawal functionality	Integration Testing
BankAccount constructor validation	
TransactionService deposit processing	
TransactionService withdrawal processing	
NotificationService alert verification	

Test Approach

- Testing Type: Unit Testing
- Testing Method: Automated
- Framework: JUnit 5
- Mocking Library: Mockito 4
- Build Tool: Maven
- Pattern: AAA (Arrange, Act, Assert)

Test Environment

- IDE: VS Code
- Language: Java 8
- Operating System: Windows
- Testing Framework: JUnit Jupiter 5.9.3
- Mocking Framework: Mockito 4.11.0

Entry Criteria

- Source code compiles with no errors
- Maven dependencies are downloaded
- JDK 8 is installed
- Test environment is configured

Exit Criteria

- All test cases have been executed
- All tests pass with zero failures
- All exceptions are correctly thrown
- Mock interactions are verified

Test Case Summary

Bank Account Test

Test ID	Class	Method	Description
TC001	BankAccountTest	valid_withdrawal_decreases_balance	A valid withdrawal decreases balance
TC002	BankAccountTest	withdrawal_moreThanBalance_throwsException	Overdraft throws an exception.
TC003	BankAccountTest	zero_withdrawalAmount_throwsException	Zero withdrawal throws an exception.
TC004	BankAccountTest	negative_withdrawalAmount_throwsException	Negative withdrawal throws an exception.
TC005	BankAccountTest	maximum_withdrawal_amount	Withdraw exact balance leaves zero
TC006	BankAccountTest	valid_deposit_increases_balance	A valid deposit increases balance
TC007	BankAccountTest	deposit_negativeAmount_throwsException	A negative deposit throws an exception.
TC008	BankAccountTest	zero_depositAmount_throwsException	Zero deposit throws an exception.
TC009	BankAccountTest	bankAccount_with_negativeBalance	A negative initial balance throws an exception.
TC010	BankAccountTest	constructor_zeroBalance_isValid	Zero initial balance is valid

Transaction Service Test

Test ID	Class	Method	Description
TC011	TransactionServiceTest	processDeposit_validAmount_increasesBalance	Deposit updates balance
TC012	TransactionServiceTest	processDeposit_sendsNotification	Deposit triggers alert
TC013	TransactionServiceTest	processWithdrawal_validAmount_decreasesBalance	Withdrawal updates balance
TC014	TransactionServiceTest	processWithdrawal_sendsNotification	Withdrawal triggers alert
TC015	TransactionServiceTest	processWithdrawal_insufficientFunds_noNotificationSent	Insufficient funds, no alert sent

User Login Test

Test ID	Class	Method	Description
TC016	UserLoginTest	valid_login_credentials_returnTrue	Valid credentials return true
TC017	UserLoginTest	wrongPassword_returnFalse	Wrong password returns false
TC018	UserLoginTest	wrongUsername_returnsFalse	Wrong username returns false
TC019	UserLoginTest	emptyUserName_throwsException	Empty username throws exception
TC020	UserLoginTest	emptyPassword_throwsException	Empty password throws exception
TC021	UserLoginTest	passwordTooShort_throwsException	Password under 6 chars throws exception
TC022	UserLoginTest	nullUserName_throwsException	Null username throws exception
TC023	UserLoginTest	nullPassword_throwsException	Null password throws exception